Quantitative Methods for Business and Economics

Current Syllabus Outline

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Course description

This course introduces the mathematical tools most commonly used in business, economics, and related applied social-science settings. The emphasis is on understanding how mathematical expressions represent real-world relationships, and on using those tools to solve practical problems involving growth, comparison, graphing, and optimization.

The course begins with algebraic foundations, then develops equation systems and graphical reasoning, followed by logarithms and growth models, and concludes this current portion with an introduction to single-variable calculus and optimization.

Learning goals

By the end of this portion of the course, students should be able to:

- work confidently with algebraic expressions, percentages, ratios, and equations;
- interpret and graph linear, quadratic, and simple nonlinear relationships;
- solve systems of equations and explain their graphical meaning;
- use logarithms to solve exponential-growth problems and interpret log scales;
- understand derivatives as rates of change and marginal effects;
- apply basic optimization tools to simple business and economics examples.

Topics and sequence

Module 1. Foundations for quantitative reasoning

- Order of operations
- Variables and mathematical notation
- Greek letters commonly used in quantitative work
- Percentages, proportions, and percentage change
- Ratios
- Basic equations, functions, and inequalities

Module 2. Algebra for business and economics

- Real numbers

- Summation and sigma notation
- Averages
- Exponents and exponent rules
- Radicals
- Quadratic equations and roots
- Absolute value

Module 3. Equation systems and graphing

- Solving systems of equations
- The Cartesian plane and plotting points
- Linear equations, slope, and intercepts
- Graphing lines and equation systems
- Parallel lines, coincident lines, and no-solution cases
- Quadratic and cubic functions

Module 4. Logarithms, growth, and scaling

- Definition of logarithms
- Properties of logarithms
- Solving logarithmic and exponential equations
- Exponential growth
- Compound growth and doubling-time calculations
- Continuous growth with e
- Base-10 and natural-log scales
- Interpretation of logarithmic transformations in business and economics

Module 5. Single-variable calculus and optimization

- Rates of change and tangent lines
- Derivatives and derivative notation
- Power, sum, product, quotient, and chain rules
- Marginal interpretation of derivatives
- Critical points
- Local maxima and minima
- Second-derivative test
- Introductory optimization notation

Representative course examples

Examples used in this portion of the course include:

- pricing, discounts, tax-inclusive prices, and percentage changes;
- break-even style algebraic reasoning;
- graphing supply-and-demand-type relationships;
- compound interest and continuous growth;
- marginal profit in a simple car-sales example;
- optimization of a single-variable objective function.

Expected mathematical emphasis

The course emphasizes intuition and interpretation alongside symbolic manipulation. Students are expected to:

- show algebraic steps clearly;
- connect formulas to graphs and economic meaning;
- interpret rates of change and growth in plain language;
- use mathematics as a tool for business and economics reasoning rather than as a purely abstract exercise.

Notes on current structure

This syllabus reflects the current active course sequence based on the slide materials available in the repository up to the present calculus and optimization lecture.